SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	**Course Title:**	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	**Assessment:**	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	**CO1:** *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*		
Student Name:	G Vishnu Madhav	**Reg. No.:**	192425192
Section / Batch:	AIML - 2024	**Date:**	27-7-26

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.
Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.
For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.
No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
		GRAND TOTAL:	**100 Marks**

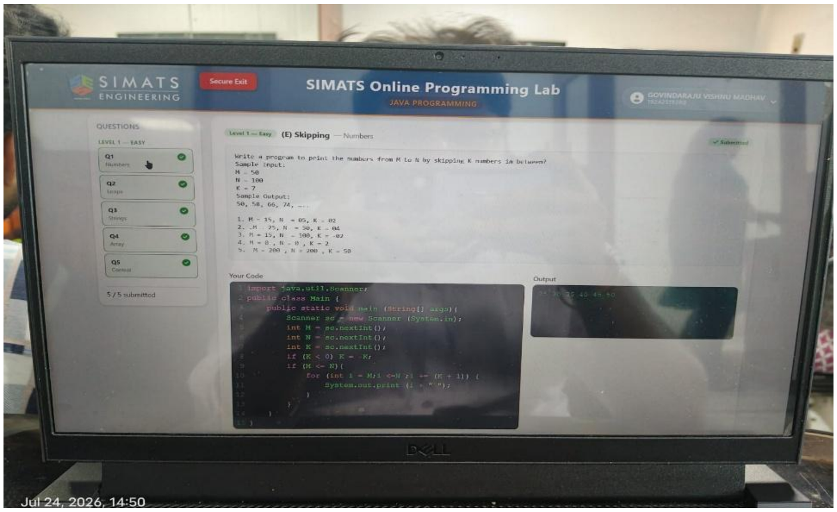
SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing)   [20 Marks]

1. Prime Numbers from m to n (skip k)

```
int m=10,n=30,k=17;
for(int i=m;i<=n;i++){
   if(i==k) continue;
   int c=0;
   for(int j=2;j<=i/2;j++)
      if(i%j==0) c++;
```

Answer / Program with output:

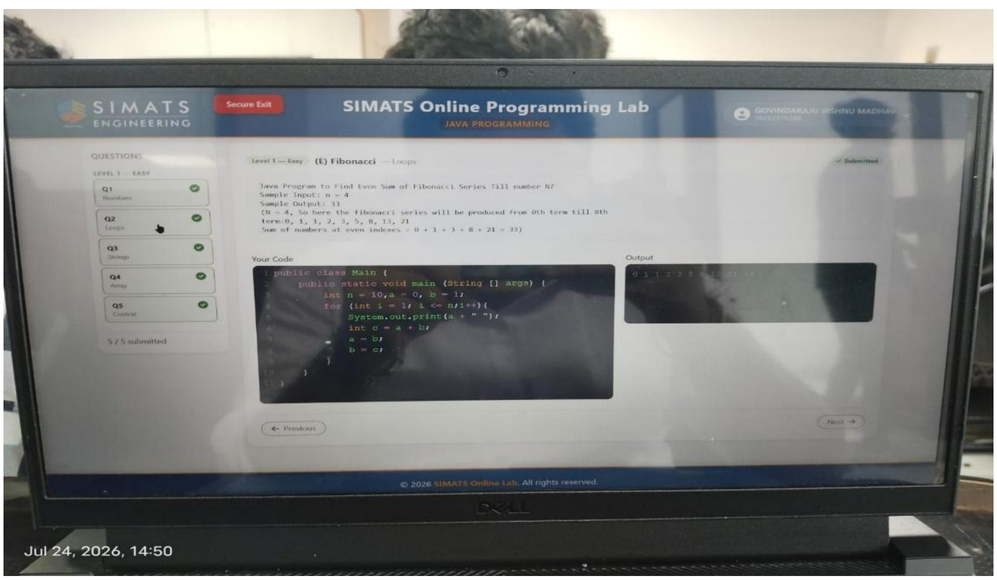


Q2. (Level 1 – Program Writing) [20 Marks]

2. Fibonacci Series

```
class Fib {
    public static void main(String[] args) {
        int n=5,a=0,b=1;
        for(int i=1;i<=n;i++){
            System.out.print(a+" ");
```

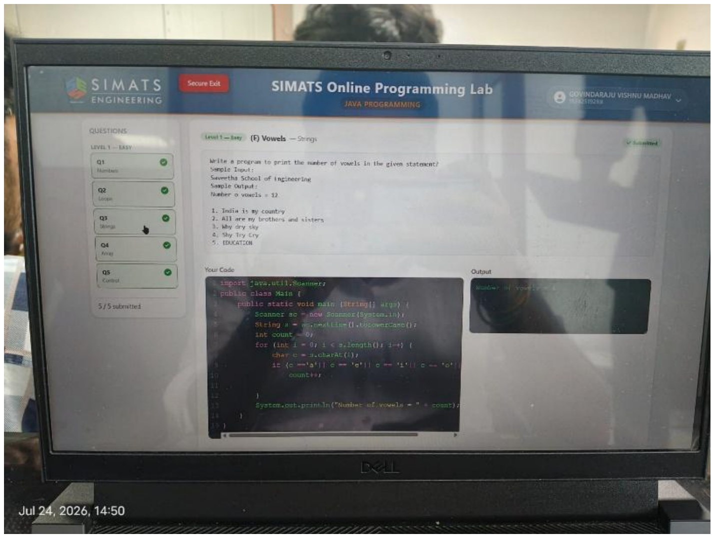
Answer / Program with output:



Q3. (Level 1 – Program Writing) [20 Marks]

```
class Vowels {
    public static void main(String[] args) {
        String s="hello";
        for(char c:s.toCharArray())
            if("aeiouAEIOU".indexOf(c)!=-1)
                System.out.print(c+" ");
    }
}
```

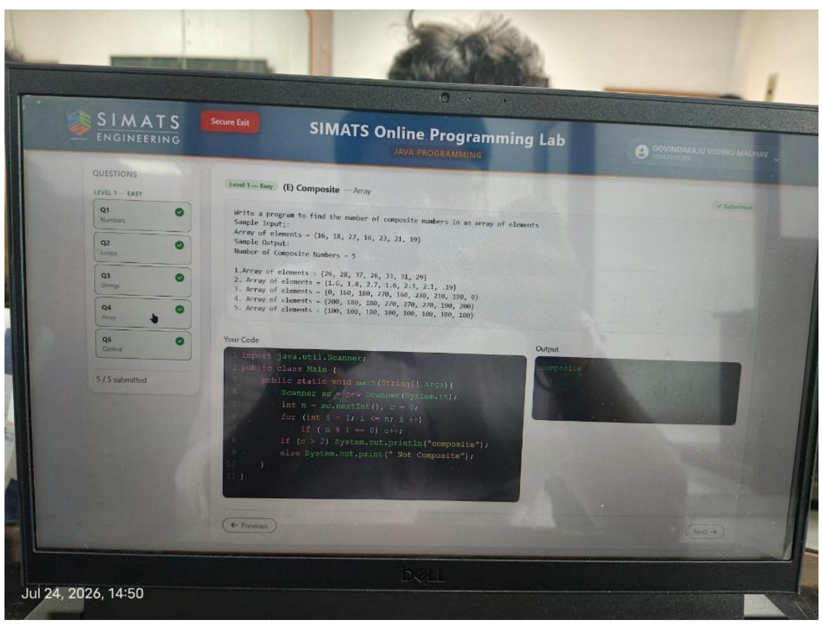
Answer / Program with output:



Q4. (Level 1 – Program Writing) [20 Marks]

```
class Composite {
  public static void main(String[] args) {
    int n=10,c=0;
    for(int i=1;i<=n;i++)
      if(n%i==0) c++;
    if(c>2) System.out.println("Composite");
    else System.out.println("Not Composite");
  }
```

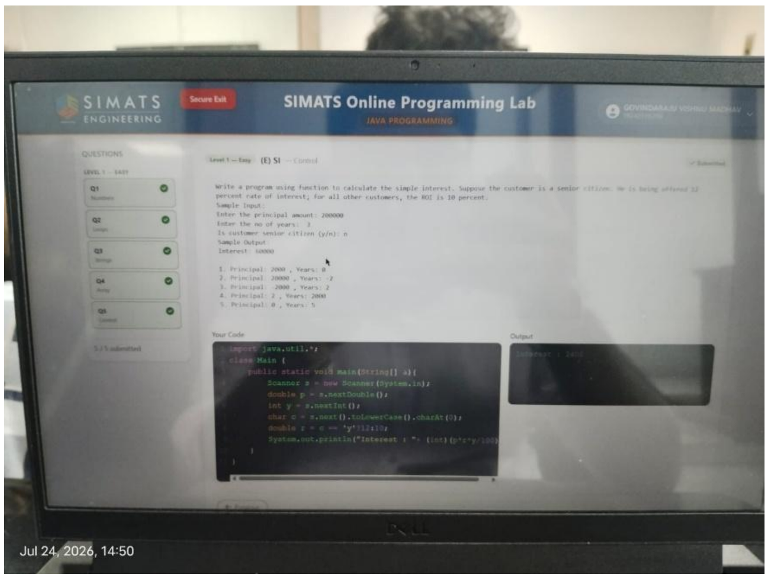
Answer / Program with output:



Q5. (Level 1 – Program Writing)   **[20 Marks]**

```
class SI {
   public static void main(String[] args) {
      double p=1000,r=5,t=2;
      System.out.println((p*r*t)/100);
   }
}
```

Answer / Program with output:



RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I G Vishnu Madhav (192425192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: GVishnuMadhav